The Two-Site Inclusion Process: Complete Hahn Spectrum and the Absorbing Face

HCS-C326 theorem-and-evidence package

3 September 2026 — revision round 2

Abstract

We diagonalize the convention-locked two-site symmetric inclusion process. The result includes its beta-binomial law, every Hahn mode and finite-sum norm, the full semigroup kernel, sharp L^2 decay, and the singular zero-parameter absorbing limit. Exact rational grids audit, but do not prove, the theorem.

1 Frozen generator and reversible law

Fix $N \in \mathbb{Z}_{\geq 0}$ and $\alpha > 0$. State $x \in S_N = \{0, \dots, N\}$ is the first-site occupancy; the second is $N - x$. Our unscaled continuous-time generator is

$$(Lf)(x) = b_x[f(x+1) - f(x)] + d_x[f(x-1) - f(x)], \quad b_x = (N-x)(\alpha + x), \quad d_x = x(\alpha + N - x), \tag{1}$$

with unavailable endpoint terms zero. Put

$$\pi(x) = \frac{N!(\alpha)_x(\alpha)_{N-x}}{x!(N-x)!(2\alpha)_N}. \tag{2}$$

Theorem 1. *For $N \geq 1$, (2) is the unique reversible law. Define*

$$H_j(x) = {}_3F_2\left(\begin{matrix} -j, j+2\alpha-1, -x \\ \alpha, -N \end{matrix}; 1\right), \quad h_j = \sum_{x=0}^{N} \pi(x) H_j(x)^2, \quad 0 \leq j \leq N.$$

Then $h_j > 0$, the H_j are a complete orthogonal basis, and

$$LH_j = -\lambda_j H_j, \qquad \lambda_j = j(j-1+2\alpha).$$

The complete transition kernel is

$$p_t(x, y) = \pi(y) \sum_{j=0}^{N} e^{-\lambda_j t} \frac{H_j(x) H_j(y)}{h_j}. \tag{3}$$

For centered f, $\|e^{tL} f\|_{2,\pi} \leq e^{-2\alpha t} \|f\|_{2,\pi}$; the exponent is sharp. At $\alpha = 0$, the endpoints absorb, absorption is almost sure, and $\mathbb{P}_x(X_\infty = N) = x/N$ for $N \geq 1$. Moreover $\{c\delta_0 + (1-c)\delta_N : 0 \leq c \leq 1\}$ is the complete stationary family, and $\pi_{\alpha,N} \Rightarrow (\delta_0 + \delta_N)/2$ as $\alpha \downarrow 0$. For $N = 0$ the process and law are the singleton.

The adjacent identity $\pi(x) b_x = \pi(x+1) d_{x+1}$ follows by cancelling Pochhammer factors; Chu–Vandermonde gives the denominator in (2). For $N \geq 1$ all interior edge rates are positive, proving irreducibility, uniqueness, and self-adjointness in $L^2(\pi)$.

The evidence grid uses $\alpha \in \{1/2, 1, 3/2, 2\}$ and $0 \leq N \leq 8$: 36 parameter rows, 180 states, and 180 full Hahn vectors. All entries are exact rationals independently reconstructed; finite evidence is not a proof.

2 Full Hahn diagonalization

Lemma 1. *The degree-j polynomial filtration is invariant under L, and the diagonal entry on its one-dimensional quotient is $-j(j-1+2\alpha)$. The terminating series defining H_j has degree j, satisfies $LH_j = -j(j-1+2\alpha)H_j$, and obeys $H_j(0) = 1$.*

Proof. For $f(x) = x^j$, the prospective x^{j+1} terms in (1) cancel. The x^j coefficient is

$$j[(N-\alpha) - (N+\alpha)] - j(j-1) = -j(j-1+2\alpha).$$

For identification, write the finite expression

$$H_j(x) = \sum_{k=0}^{j} \frac{(-j)_k (j+2\alpha-1)_k (-x)_k}{(\alpha)_k (-N)_k k!}.$$

Put $\phi_k(x) = (-x)_k$. Direct forward and backward differencing gives

$$L\phi_k = -k(k-1+2\alpha)\phi_k - k(N-k+1)(\alpha+k-1)\phi_{k-1}.$$

For $H_j = \sum_k c_k \phi_k$, the eigenvalue equation requires

$$\frac{c_{k+1}}{c_k} = \frac{\lambda_j - \lambda_k}{(k+1)(N-k)(\alpha+k)} = \frac{(j-k)(j+k+2\alpha-1)}{(k+1)(N-k)(\alpha+k)}.$$

This is exactly the ratio of consecutive coefficients in the displayed terminating sum, by $(a)_{k+1} = (a+k)(a)_k$. Consequently every coefficient on the left of

$$b_x[H_j(x+1) - H_j(x)] + d_x[H_j(x-1) - H_j(x)] + j(j-1+2\alpha)H_j(x)$$

is zero. This proves the finite difference identity. Its denominators do not vanish for $j \le N$ and $\alpha > 0$; its final coefficient is nonzero, while substitution $x = 0$ gives one. $\square$

Because $\lambda_{j+1} - \lambda_j = 2(j+\alpha) > 0$, these are all the distinct diagonal values from Lemma 1. Self-adjointness makes their eigenvectors orthogonal. There are $N+1$ of them, hence they are complete, and each displayed finite-sum norm h_j is positive. Expanding any function in this normalized basis proves (3). Parseval on the constants' orthogonal complement gives the stated bound because $\lambda_1 = 2\alpha$; $H_1(x) = 1 - 2x/N$ gives equality and therefore sharpness. When $N = 1$, this says exactly that the spectrum is $\{0, 2\alpha\}$ and $\pi = (1/2, 1/2)$.

Revision certificate: complete Hahn spectrum and semigroup

This round adds polynomial triangularity, the terminating difference identity, orthogonality with positive finite-sum norms, the full kernel, and sharp decay.

3 The singular face, ownership, and Route A

At $\alpha = 0$, $b_x = d_x = x(N-x)$. Thus both endpoints absorb and, from an interior state, the embedded chain is the finite simple symmetric walk. Absorption is almost sure. Since $Lx = 0$, bounded optional stopping yields $x = N\mathbb{P}_x(X_\infty = N)$. A stationary law cannot charge the transient interior: stationarity makes that mass time-independent, whereas finite-state almost-sure absorption makes it tend to zero. Thus all and only stationary laws are $c\delta_0 + (1-c)\delta_N$, $0 \le c \le 1$. For fixed $N \ge 1$, each endpoint numerator in (2) is asymptotic to α/N, the denominator to $2\alpha/N$, and every interior numerator is $O(\alpha^2)$; this proves the weak limit. For $N = 0$ only δ_0 exists. At $N = 1$ the hitting formula remains valid.

The inclusion model is sourced to Giardina–Redig–Vafayi [1]; the Hahn normalization, representation, and difference equation are checked against DLMF Sections 18.19, 18.20.5, and 18.22(ii) [2]. We claim no priority. C253 owns Moran fixation, C263 a growing Polya urn, C285 Gordon–Newell product form, and C322 Kac sphere collisions; none owns this theorem. We assert no multisite, open-boundary, or condensation-scaling result.

The Route-A tuple is (FAIL, FAIL, FAIL, FAIL, FORMAL HINT), with verdict `ROUTE_A_REJECTED`; Route B is false. The final entry records only a finite self-adjoint spectral analogy, not a Hilbert–Polya operator. Under `NO_BAD_EULER_OR_ROOT_NUMBER`, no target local datum, Euler factor, root number, automorphy, divisor, functional equation, or zero match is asserted.

Revision certificate: absorbing face and scope closure

This final round proves absorption, hitting probabilities, the stationary weak limit, all degenerate cases, source and collision ownership, and the firewall.

References

[1] C. Giardina, F. Redig, and K. Vafayi, Correlation inequalities for interacting particle systems with duality, *J. Stat. Phys.* 141 (2010); DOI: 10.1007/s10955-010-0055-0; arXiv:0906.4664.

[2] NIST Digital Library of Mathematical Functions, Sections 18.19, 18.20.5, and 18.22(ii), Hahn definitions, representation, and difference equations.